

PRACTICE SHEET

- What is the area under the curve $y = |x| + |x - 1|$ between $x = 0$ and $x = 1$?
(a) $1/2$ (b) 1
(c) $3/2$ (d) 2
- What is the area enclosed by the curve $2x^2 + y^2 = 1$?
(a) 2π (b) π
(c) $\pi/2$ (d) $\pi/\sqrt{2}$
- What is the area of the region bounded by the line $3x - 5y = 15$, $x = 1$, $x = 3$ and x -axis in sq unit?
(a) $36/5$ (b) $18/5$
(c) $9/5$ (d) $3/5$
- What is the area of the ellipse $4x^2 + 9y^2 = 1$?
(a) 6π (b) $\pi/36$
(c) $\pi/6$ (d) $\pi/\sqrt{6}$
- What is the value of k if the area bounded by the curve $y = \sin kx$, $y = 0$, $x = \pi / (3k)$ is 3 sq unit?
(a) $1/2$ (b) 1
(c) $3/2$ (d) 2
- What is the area bounded by the curve $y = 4x - x^2 - 3$ and the x -axis?
- What is the area under the curve $f(x) = xe^x$ above the X -axis and between the lines $x = 0$ and $x = 1$?
(a) $1/2$ sq. unit (b) 1 sq. unit
(c) $3/2$ sq. units (d) 2 sq. units
- If $f(x) = 1 - \frac{x^2}{4}$, $x \in [-2, 2]$, then find the area covered by X -axis.
(a) $8/3$ sq. units (b) $4/3$ sq. units
(c) $2/3$ sq. unit (d) $1/3$ sq. unit
- What is the area bounded by the curve $y = x^2$ and the line $y = 16$?
(a) $32/3$ (b) $64/3$
(c) $256/3$ (d) $128/3$
- What is the area enclosed between the curve $y^2 = 12x$ and the lines $x = 0$ and $y = 6$?
(a) 2 sq. units (b) 4 sq. units
(c) 6 sq. units (d) 8 sq. units

ANSWER KEYS

1.	b	2.	d	3.	b	4.	c	5.	a	6.	b	7.	d	8.	a	9.	c	10.	c
----	---	----	---	----	---	----	---	----	---	----	---	----	---	----	---	----	---	-----	---

Solutions

Sol.1. (b)
 $|x|$ for $x \geq 0$
 $= x$ and $|x - 1|$ for $x \leq 1$
 $= -(x - 1)$,

So, $\int_0^1 (|x| + |x - 1|) dx = \text{required area}$

$$a = \int_0^1 x dx - \int_0^1 (x - 1) dx$$

$$= \left[\frac{x^2}{2} \right]_0^1 - \left[\frac{x^2}{2} - x \right]_0^1 = \frac{1}{2} - \left(\frac{1}{2} - 1 \right) = 1 \text{ sq units}$$

Sol.2. (d)

Given equation of curve $2x^2 + y^2 = 1$ is an ellipse

which can be written as $\frac{x^2}{\frac{1}{2}} + \frac{y^2}{1} = 1$

Area of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $A = \pi ab$ sq unit

Here, $a = \frac{1}{\sqrt{2}}$, $b = 1$

$\therefore$ Required area $= \pi \cdot \frac{1}{\sqrt{2}} \cdot 1 = \frac{\pi}{\sqrt{2}}$ sq unit

Sol.3. (b)

The given equation of line can be rewritten as

$$\frac{x}{5} - \frac{y}{3} = 1 \text{ and } t = \frac{3x - 15}{5}$$

$\therefore$ Required area $= \int_1^3 y dx$

$$= \int_1^3 \left(\frac{3x - 15}{5} \right) dx = \frac{1}{5} \int_1^3 (3x - 15) dx$$

$$= \frac{1}{5} \left[\frac{3x^2}{2} - 15x \right]_1^3 = \frac{1}{5} \left[\frac{27}{2} - 45 - \frac{3}{2} + 15 \right]$$

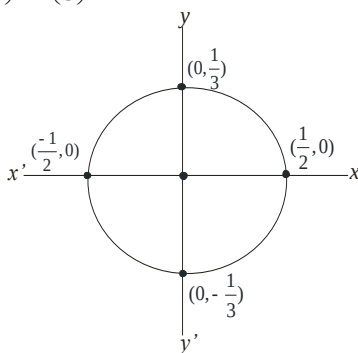
$$= \frac{1}{5} \left[\frac{24}{3} - 30 \right] = \frac{1}{5} [12 - 30]$$

$$= \frac{-18}{5} = \frac{18}{5} \text{ sq. unit (neglecting -ve sign)}$$

Sol.4. (c)

Given $4x^2 + 9y^2 = 1$

$$\frac{x^2}{\left(\frac{1}{2}\right)^2} + \frac{y^2}{\left(\frac{1}{3}\right)^2} = 1$$



$\therefore a = 1/2$ and $b = 1/3$

Now, area of ellipse $= 4 \int_0^{1/2} y dx$

$$= 4 \int_0^{1/2} \sqrt{\frac{1 - 4x^2}{9}} dx = \frac{4}{3} \int_0^{1/2} \sqrt{1 - (2x)^2} dx$$

$$= \frac{2}{3} \int_0^1 \sqrt{1 - t^2} dt = \frac{2}{3} \left[\frac{1}{2} t \sqrt{1 - t^2} + \frac{1}{2} \sin^{-1}(t) \right]_0^1$$

$$= \frac{2}{3} \left[0 + \frac{1}{2} \sin^{-1}(1) - \frac{1}{2} \sin^{-1}(0) \right]$$

$$= \frac{2}{3} \left[\frac{1}{2} \times \frac{\pi}{2} - \frac{1}{2} \times 0 \right] = \frac{\pi}{6}$$

Sol.5. (a)

Given $x = \pi/k$, $x = \frac{\pi}{3k}$

and $y = \sin kx$

Let the required area be A

So, $A = 3$ (Given)

Therefore Area, $A = \int_{\pi/3k}^{\pi/k} \sin kx dx$

$$\Rightarrow 3 = - \left[\frac{\cos kx}{k} \right]_{\pi/3k}^{\pi/k}$$

$$\Rightarrow 3 = -1/k \left[\cos \pi - \cos \frac{\pi}{3} \right]$$

$$\Rightarrow 3 = -1/k \left[-1 - \frac{1}{2} \right]$$

($\because \cos \pi = \cos(\pi/2 + \pi/2)$)

$= -\sin \pi/2$ and $\cos 60^\circ = 1/2$)

Sol.6. (b)

Given curve is $y = 4x - x^2 - 3$

Since, area bounded by x - axis

$\therefore y = 0$

$$\Rightarrow 4x - x^2 - 3 = 0 \Rightarrow x^2 - 4x + 3 = 0$$

$$\Rightarrow x^2 - 3x - x + 3 = 0 \Rightarrow (x - 3)(x - 1) = 0$$

$$\Rightarrow x = 1, 3$$

$$\therefore \text{required} = \int_1^3 (4x - x^2 - 3) dx$$

$$= \left[\frac{4x^2}{2} - \frac{x^3}{3} - 3x \right]_1^3 = \left(\frac{36}{2} - \frac{27}{3} - 9 \right) - \left(\frac{4}{2} - \frac{1}{3} - 3 \right)$$

$$= (18 - 9 - 9) - \left(2 - \frac{10}{3} \right) = 0 - \left(\frac{-4}{3} \right) = \frac{4}{3} \text{ sq. unit}$$

Sol.7. (b)

$$\text{Area} = \int_0^1 (xe^x) dx = \left[e^x (x - 1) \right]_0^1 = 1 \text{ sq. unit}$$

Sol.8. (a)

$$\text{Area} = \int_{-2}^2 \left(1 - \frac{x^2}{2} \right) dx = \left[x - \frac{x^3}{6} \right]_{-2}^2 = \frac{8}{3} \text{ sq unit}$$

Sol.9. (c)

$$\text{Area} = 2 \int_0^{16} \sqrt{y} dy = 2 \left[\frac{y^{3/2}}{3/2} \right]_0^{16} = \frac{256}{3} \text{ sq unit}$$

Sol.10. (c)

$$\text{Area} = \int_0^6 \frac{y^2}{12} dy = \left[\frac{y^3}{36} \right]_0^6 = 6 \text{ sq unit}$$

NDA PYQ

1. What is the area bounded by the curves $y = e^x$, $y = e^{-x}$ and the straight line $x = 1$?
 (a) $\left(e + \frac{1}{e}\right)$ sq. units (b) $\left(e - \frac{1}{e}\right)$ sq. units
 (c) $\left(e + \frac{1}{e} - 2\right)$ sq. units (d) $\left(e - \frac{1}{e} - 2\right)$ sq. units
[NDA (I) - 2011]
 2. What is the area bounded by the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ ($x, y \geq 0$) and the coordinate axes?
 (a) $\frac{5a^2}{6}$ (b) $\frac{a^2}{3}$
 (c) $\frac{a^2}{2}$ (d) $\frac{a^2}{6}$
[NDA (II) - 2011]
 3. What is the area between the curve $y = \cos 3x$, $0 \leq x \leq \pi/6$ and the coordinate axes?
 (a) 1 sq. unit (b) $1/2$ sq. unit
 (c) $1/3$ sq. unit (d) $1/4$ sq. unit
[NDA (II) - 2011]
 4. What is the area enclosed by the equation $x^2 + y^2 = 2$?
 (a) 4π sq. units (b) 2π sq. units
 (c) $4\pi^2$ sq. units (d) 4 sq. units
[NDA (II) - 2011]
 5. What is the area of the triangle formed by the lines joining the vertex of the parabola $x^2 = 12y$ to the ends of the latus rectum?
 (a) 9 square units (b) 12 square units
 (c) 14 square units (d) 18 square units
[NDA-2011(2)]
 6. The area bounded by the curve $x = f(y)$, the Y-axis and the two lines $y = a$ and $y = b$ is equal to:
 (a) $\int_a^b y \, dx$ (b) $\int_a^b y^2 \, dx$
 (c) $\int_a^b x \, dy$ (d) None of these
[NDA (I) - 2012]
 7. What is the area of the region bounded by the lines $y = x$, $y = 0$ and $x = 4$?
 (a) 4 sq. units (b) 8 sq. units
 (c) 12 sq. units (d) 16 sq. units
[NDA (II) - 2012]
 8. What is the area of the portion of the curve $y = \sin x$, lying between $x = 0$, $y = 0$ and $x = 2\pi$?
 (a) 1 sq. unit (b) 2 sq. units
 (c) 4 sq. units (d) 8 sq. units
[NDA (II) - 2012]
 9. What are the area of the region enclosed by $y = 2|x|$ and $y = 4$?
 (a) 2 sq. units (b) 4 sq. units
 (c) 8 sq. units (d) 16 sq. units
[NDA (I) - 2013]
 10. What is the area of parabola $y^2 = x$ bounded by its latus rectum?
 (a) $1/12$ sq. unit (b) $1/6$ sq. unit
 (c) $1/3$ sq. unit (d) None of the above
[NDA (I) - 2013]
 11. What is the area bounded by the lines $x = 0$, $y = 0$ and $x + y + 2 = 0$?
 (a) $1/2$ sq. unit (b) 1 sq. unit
 (c) 2 sq. units (d) 4 sq. units
[NDA (I) - 2013]
 12. What is the area bounded by the lines $x = 0$, $y = 0$ and $x + y + 2 = 0$?
 (a) $1/2$ sq. unit (b) 1 sq. unit
 (c) 2 sq. units (d) 4 sq. units
[NDA (II) - 2013]
 13. What is the area of the parabola $x^2 = y$ bounded by the line $y = 1$?
 (a) $1/3$ sq. unit (b) $2/3$ sq. unit
 (c) $4/3$ sq. units (d) 2 sq. units
[NDA (II) - 2013]
 14. What is the area bounded by $y = \tan x$, $y = 0$ and $x = \pi/4$?
 (a) $\log 2$ sq. units (b) $\frac{\log 2}{2}$ sq. units
 (c) $2(\log 2)$ sq. units (d) None of these
[NDA (II) - 2013]
- For next three (03) items that follow:**
 The line $2y = 3x + 12$ cuts the parabola $4y = 3x^2$
15. Where does the line cut the parabola?
 (a) At $(-2, 3)$ only
 (b) At $(4, 12)$ only
 (c) At both $(-2, 3)$ and $(4, 12)$
 (d) Neither at $(-2, 3)$ nor at $(4, 12)$
[NDA (II) - 2014]
 16. What is the area enclosed by the parabola and the line?
 (a) 27 square unit (b) 36 square unit
 (c) 48 square unit (d) 54 square unit
[NDA (II) - 2014]
 17. What is the area enclosed by the parabola, the line and the y-axis in the first quadrant?
 (a) 7 square unit (b) 14 square unit
 (c) 20 square unit (d) 21 square unit
[NDA (II) - 2014]
 18. What are the area of the parabola $y^2 = 4bx$ bounded by its latus rectum?
 (a) $\frac{2b^2}{3}$ sq. units (b) $\frac{4b^2}{3}$ sq. units
 (c) b^2 sq. units (d) $\frac{8b^2}{3}$ sq. units
[NDA (II) - 2014]
- Directions (for next two):** Consider the line $x = \sqrt{3}y$ and the circle $x^2 + y^2 = 4$
19. What is the area of the region in the first quadrant enclosed by the X-axis, the line $x = \sqrt{3}$ and the circle?
 (a) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$ (b) $\frac{\pi}{2} - \frac{\sqrt{3}}{2}$

- (c) $\frac{\pi}{3} - \frac{1}{2}$ (d) None of the above

[NDA (I) - 2015]

20. What is the area of the region in the first quadrant enclosed by the X-axis, the line $y = \sqrt{3}x$ and the circle?

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$
(c) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$ (d) None

[NDA (I) - 2015]

Directions (for next two): Consider the curves $y = \sin x$ and $y = \cos x$

21. What is the area of the region bounded by the above two curves and the lines $x=0$ and $x=\pi/4$?

- (a) $\sqrt{2} - 1$ (b) $\sqrt{2} + 1$
(c) $\sqrt{2}$ (d) 2

[NDA (I) - 2015]

22. What is the area of the region bounded by the above two curves and the lines $x=\pi/4$ and $x=\pi/2$?

- (a) $\sqrt{2} - 1$ (b) $\sqrt{2} + 1$
(c) $2\sqrt{2}$ (d) 2

[NDA (I) - 2015]

23. The area bounded by the coordinate axes and the curve $\sqrt{x} + \sqrt{y} = 1$, is :

- (a) 1 square units (b) $1/2$ square units
(c) $1/3$ square units (d) $1/6$ square units

[NDA (II) - 2015]

For the next two (2) items that follow:

Consider the curves

$$y = |x-1| \text{ and } |x| = 2$$

24. What is / are the point(s) of intersection of the curves?

- (a) Only $(-2, 3)$
(b) Only $(2, 1)$
(c) $(-2, 3)$ and $(2, 1)$
(d) Neither $(-2, 3)$ nor $(2, 1)$

[NDA (I) - 2016]

25. What is the area of the region bounded by the curves and x-axis?

- (a) 3 square units (b) 4 square units
(c) 5 square units (d) 6 square units

[NDA (I) - 2016]

For the next two (2) items that follow

Consider the curves

$$f(x) = x|x| - 1 \text{ and } g(x) = \begin{cases} \frac{3x}{2}, & x > 0 \\ 2x, & x \leq 0 \end{cases}$$

26. Where do the curves intersect?

- (a) At $(2, 3)$ only
(b) At $(-1, -2)$ only
(c) At $(2, 3)$ and $(-1, -2)$
(d) Neither at $(2, 3)$ nor at $(-1, -2)$

[NDA (I) - 2016]

27. What is the area bounded by the curves?

- (a) $17/6$ square units (b) $8/3$ square units
(c) 2 square (d) $1/3$ square units

[NDA (I) - 2016]

Direction for two questions:

Given function $f(x) = |x - 1| + x^2$ where $x \in \mathbb{R}$.

28. What is the area of the region bounded by x-axis, the curve, $y = f(x)$ and the two ordinates $x = 1/2$ and $x = 1$?

- (a) $5/12$ square unit (b) $5/6$ square unit
(c) $7/6$ square unit (d) 2 square units

[NDA (I) - 2016]

29. What is the area of the region bounded by x-axis, the curve $y = f(x)$ and the two ordinates $x = 1$ and $x = 3/2$?

- (a) $5/12$ square unit (b) $7/12$ square unit
(c) $2/3$ square unit (d) $11/12$ square unit

[NDA (I) - 2016]

30. What is the area bounded by the curves $|y| = 1 - x^2$?

- (a) $4/3$ square units (b) $8/3$ square units
(c) 4 square units (d) $16/3$ square units

[NDA (II) - 2016]

31. The area bounded by the curve $|x| + |y| = 1$ is:

- (a) 1 square unit (b) $2\sqrt{2}$ square unit
(c) 2 square units (d) $2\sqrt{3}$ square units

[NDA (II) - 2017]

32. What is the area of the region bounded by the parabola $y^2 = 6(x-1)$ and $y^2 = 3x$?

- (a) $\frac{\sqrt{6}}{3}$ (b) $\frac{2\sqrt{6}}{3}$
(c) $\frac{4\sqrt{6}}{3}$ (d) $\frac{5\sqrt{6}}{3}$

[NDA (I) - 2018]

33. What is the area of one of the loops between the curve $y = c \sin x$ and x-axis?

- (a) c (b) 2c
(c) 3c (d) 4c

[NDA (I) - 2019]

34. What is the area of the region bounded by $|x| < 5$, $y = 0$ and $y = 8$?

- (a) 40 sq unit (b) 80 sq unit
(c) 120 sq unit (d) 160 sq unit

[NDA (II) - 2019]

35. What is the area of the region enclosed between the curve $y^2 = 2x$ and the straight line $y=x$?

- (a) $2/3$ sq unit (b) $4/3$ sq unit
(c) $1/3$ sq unit (d) 1 sq unit

[NDA (II) - 2019]

36. What is the area of the region enclosed between the curve $y^2 = 2x$ and straight line $y = x$?

- (a) $1/2$ (b) 1
(c) $2/3$ (d) 2

[NDA - 2020]

37. What is the area bounded by $y = \sqrt{16 - x^2}$, $y \geq 0$ and x-axis?

- (a) 16π sq units (b) 8π sq units
(c) 4π sq units (d) 2π sq units

[NDA (I) - 2021]

38. What is the area bounded by $y = [x]$, where $[.]$ is the greatest integer function, the x-axis and the lines $x = -1.5$ and $x = -1.8$?

- (a) 0.3 square unit (b) 0.4 square unit
(c) 0.6 square unit (d) 0.8 square unit

[NDA (II) - 2021]

39. What is the area included in the first quadrant between the curves $y = x$ and $y = x^3$?

- (a) $\frac{1}{8}$ square unit (b) $\frac{1}{4}$ square unit
(c) $\frac{1}{2}$ square unit (d) 1 square unit

[NDA (I) - 2022]

40. The area of the region bounded by the parabola $y^2 = 4kx$, where $k > 0$ and its latus rectum is 24 square units. What is the value of k ?

- (a) 1 (b) 2
(c) 3 (d) 4

[NDA (I) - 2022]

41. What is the area of the region enclosed in the first quadrant by $x^2 + y^2 = \pi^2$, $y = \sin x$ and $x = 0$?

- (a) $\frac{\pi^3}{4} - 1$ (b) $\frac{\pi^3}{4} - 2$
(c) $\frac{\pi^3}{2} - 1$ (d) $\frac{\pi^2}{4} - 2$

[NDA 2022 (II)]

42. What is the area of the region (in the first quadrant) bounded by $y = \sqrt{1-x^2}$, $y = x$ and $y = 0$?

- (a) $\pi/4$ (b) $\pi/6$
(c) $\pi/8$ (d) $\pi/12$

[NDA 2022 (II)]

43. What is the area of the region bounded by $x - |y| = 0$ and $x - 2 = 0$?

- (a) 1 (b) 2
(c) 4 (d) 8

[NDA 2022 (II)]

44. For what value of m with $m < 0$, is the area bounded by the lines $y = x$, $y = mx$ and $x = 2$ equal to 3?

- (a) $-1/2$ (b) -1
(c) $-3/2$ (d) -2

[NDA - 2023 (1)]

Consider the following for the next two (02) items that follow:

The line $y = x$ partitions the circle $(x-a)^2 + y^2 = a^2$ in two segments.

45. What is the area of minor segment?

- (a) $\frac{(\pi-2)a^2}{4}$ (b) $\frac{(\pi-1)a^2}{4}$
(c) $\frac{(\pi-2)a^2}{2}$ (d) $\frac{(\pi-1)a^2}{2}$

[NDA - 2023 (1)]

46. What is the area of major segment?

- (a) $\frac{(3\pi-2)a^2}{4}$ (b) $\frac{(3\pi+2)a^2}{4}$
(c) $\frac{(3\pi-2)a^2}{2}$ (d) $\frac{(3\pi+1)a^2}{2}$

[NDA - 2023 (1)]

47. What is the area between the curve $f(x) = x|x|$ and x -axis for $x \in [-1, 1]$?

- (a) $2/3$ (b) $1/2$
(c) $1/4$ (d) $1/3$

[NDA-2023 (2)]

Consider the following for the next two (02) items that follow:

The circle $x^2 + y^2 - 2x = 0$ is partitioned by line $y = x$ in two segments. Let A_1 , A_2 be the areas of major and minor segments respectively.

48. What is the value of A_1 ?

- (a) $\frac{\pi-2}{4}$ (b) $\frac{\pi+2}{4}$
(c) $\frac{3\pi-2}{4}$ (d) $\frac{3\pi+2}{4}$

[NDA-2024 (1)]

49. What is the value of $\frac{2(A_1 + A_2)}{A_1 - 3A_2}$?

- (a) π (b) 1
(c) -1 (d) $-\pi$

[NDA-2024 (1)]

Direction: Consider the following for the two items given below

The area bounded by the parabola $y^2 = kx$ and the line $x = k$, where $k > 0$, is $4/3$ square units.

50. What is the value of k ?

- (a) $1/2$ (b) 1
(c) $\sqrt{2}$ (d) 2

[NDA-2024 (2)]

51. What is the area of the parabola bounded by the latus rectum?

- (a) $1/6$ square units (b) $2/3$ square units
(c) 1 square units (d) $4/3$ square units

[NDA-2024 (2)]

Consider the following for the one (01) items that follow:

The slope of the tangent to the curve $y = f(x)$ at $(x, f(x))$ is 4 for every real number x and the curve passes through the origin.

52. What is the area bounded by the curve, the x -axis and the line $x = 4$?

- (a) 8 square units (b) 16 square units
(c) 32 square units (d) 64 square units

[NDA-2025 (1)]

Consider the following for the one (01) items that follow:

Let the curve $f(x) = |x-3|$

53. What is the area bounded by the curve $f(x)$ and $y = 3$?

- (a) 3 square units (b) 4.5 square units
(c) 7.5 square units (d) 9 square units

[NDA-2025 (1)]

Consider the following for the one (01) items that follow:

54. Let the function $f(x) = x^2 - 1$

What is the area bounded by the function $f(x)$ and the x -axis?

- (a) $1/3$ square unit (b) $2/3$ square unit
(c) $4/3$ square units (d) 2 square units

[NDA-2025 (1)]

For the following one items:

Consider the function $f(x) = x|x|$

55. What is area bounded by the curve $f(x)$, the x -axis and the lines $x = -2$ and $x = 1$?

- (a) $1/3$ (b) $2/3$
(c) $5/2$ (d) 3

[NDA-2025 (2)]

56. What is the area of the region bounded by $|x| \leq 2k$ and $|y| \leq k$, where k is a positive real number?

- (a) $2k^2$ (b) $4k^2$
(c) $5k^2$ (d) $8k^2$

[NDA-2025 (2)]

ANSWER KEY																			
1.	c	2.	d	3.	c	4.	b	5.	d	6.	c	7.	b	8.	c	9.	c	10.	b
11.	c	12.	c	13.	c	14.	b	15.	c	16.	a	17.	c	18.	d	19.	a	20.	a
21.	a	22.	a	23.	d	24.	c	25.	c	26.	c	27.	b	28.	a	29.	d	30.	b
31.	c	32.	c	33.	b	34.	b	35.	a	36.	c	37.	b	38.	c	39.	b	40.	c
41.	b	42.	c	43.	c	44.	a	45.	a	46.	b	47.	a	48.	d	49.	a	50.	b
51.	a	52.	c	53.	d	54.	c	55.	d	56.	d								

Solutions

Sol. 1. (c)

$$\text{Area} = \int_0^1 (e^x - e^{-x}) dx = [e^x + e^{-x}]_0^1 = e + \frac{1}{e} - 2$$

Sol. 2. (d)

$$\sqrt{x} + \sqrt{y} = \sqrt{a}$$

$$\text{then } \sqrt{y} = \sqrt{a} - \sqrt{x}$$

$$y = a + x - 2\sqrt{a}\sqrt{x}$$

$$\text{Area} = \int_0^a (a + x - 2\sqrt{a}\sqrt{x}) dx$$

$$\left[ax + \frac{x^2}{2} - 2\sqrt{a} \frac{x^{3/2}}{3/2} \right]_0^a = a^2 + \frac{a^2}{2} - \frac{4}{3}a^2 = \frac{a^2}{6}$$

Sol. 3. (c)

$$\text{Area} = \int_0^{\pi/6} \cos 3x dx = \left[\frac{\sin 3x}{3} \right]_0^{\pi/6} = \frac{1}{3}$$

Sol. 4. (b)

Area of circle is πr^2 here r^2 is given 2

$$\text{Area} = 2\pi$$

Sol. 5. (d)

Vertex of parabola (0,0)

End points of latus rectum ((6,3) and (-6,3))

area of triangle = $(1/2)(12)(3) = 18$ sq. unit

Sol. 6. (c)

The area bounded by the curve $x = f(y)$, the Y-axis and the two lines $y = a$ and $y = b$ is equal to

$$= \int_a^b x dy$$

Sol. 7. (d)

$$x = 0, y = 0, x = 4$$

these three lines make a right angled triangle of base and height 4, 4 respectively

so area = $1/2 \times 4 \times 4 = 8$ sq unit

Sol. 8. (c)

Required area =

$$\int_0^{\pi} \sin x dx + \left| \int_{\pi}^{2\pi} \sin x dx \right| = 2 + 2 = 4$$

Sol. 9. (c)

$$\text{Area} = 2 \int_0^4 \frac{y}{2} dy = 2 \left[\frac{y^2}{4} \right]_0^4 = 8 \text{ sq unit}$$

Sol. 10. (b)

letus rectum of parabola $y^2 = x$ is $x = 1/4$ line

$$\text{Area} = 2 \int_0^{1/4} \sqrt{x} dx = 2 \left[\frac{x^{3/2}}{3/2} \right]_0^{1/4} = \frac{1}{6} \text{ sq unit}$$

Sol. 11. (c)

$$x = 0, y = 0, x + y = 2$$

these three lines make a right angled triangle of base and height 2, 2 respectively

so area = $1/2 \times 2 \times 2 = 2$ sq unit

Sol. 12. (c)

$$x = 0, y = 0, x + y = 2$$

these three lines make a right angled triangle of base and height 2, 2 respectively

so area = $1/2 \times 2 \times 2 = 2$ sq unit

Sol. 13. (c)

$$\text{Area} = 2 \int_0^1 \sqrt{y} dy = 2 \left[\frac{y^{3/2}}{3/2} \right]_0^1 = \frac{4}{3} \text{ sq unit}$$

Sol. 14. (b)

$$\text{Area} = \int_0^{\pi/4} \tan x dx = [\log \sec x]_0^{\pi/4}$$

$$= \log \sqrt{2} = \frac{1}{2} \log 2 \text{ sq unit}$$

Sol. 15. (c)

Equation of line is $2y = 3x + 12$... (i)

Equation of parabola is

$$4y = 3x^2 \dots (ii)$$

On putting the value of y from Eq. (i) to Eq. (ii),

we get

$$2(3x + 12) = 3x^2$$

$$\Rightarrow 3x^2 - 6x - 24 = 0$$

$$\Rightarrow x^2 - 2x - 8 = 0$$

$$\Rightarrow (x-4)(x+2) = 0$$

$$\Rightarrow x = 4 \text{ and } x = -2$$

On putting these values in Eq. (ii), we get

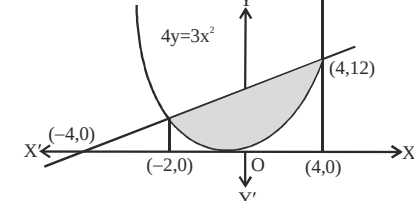
$$y = 12 \text{ and } y = 3$$

Hence, points are (-2, 3) and (4, 12)

Sol. 16. (a)

Area enclosed by the parabola and the line

$$2y = 3x + 12$$



$$= \int_{-2}^4 \left[\frac{(3x+12)}{2} - \frac{3x^2}{4} \right] dx$$

$$= \frac{1}{3} \left[\frac{3x^2}{2} + 12x \right]_{-2}^4 - \frac{3}{4} \left[\frac{x^3}{3} \right]_{-2}^4$$

$$= \frac{1}{2} \left[\frac{3(4)^2}{2} + 12(4) \right] - \left\{ \frac{3(-2)^2}{2} + 12(-2) \right\}$$

$$- \frac{3}{4} \left[\frac{4^3}{3} - \frac{(-2)^3}{3} \right]$$

$$= \frac{1}{2} [24 + 48 - 6 + 24] - \frac{3}{4} \left[\frac{64 + 8}{3} \right] = \frac{1}{2} (90) - 18$$

$$= 45 - 18 = 27 \text{ sq. units}$$

Sol. 17. (c)

Area enclosed by the parabola, the line and the Y-axis in the first quadrant.

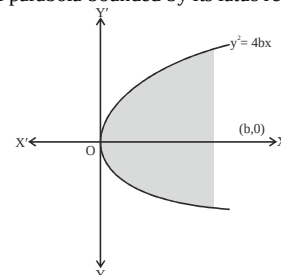
$$= \int_0^4 \left(\frac{3x+12}{2} - \frac{2x^2}{4} \right) dx = \left[\frac{3}{4} x^2 + 6x - \frac{x^3}{4} \right]_0^4$$

$$= 3 \times 4 + 24 - 16 = 12 + 24 - 16 = 20 \text{ sq.}$$

Sol. 18. (d)

Equation of parabola is $y^2 = 4bx$

Area of parabola bounded by its latus rectum



$$= 2 \int_0^4 \sqrt{4bx} dx$$

$$= 4\sqrt{b} \times \frac{2}{3} \left[x^{3/2} \right]_0^b = \frac{8\sqrt{b}}{3} [b^{3/2} - 0] = \frac{8b^2}{3}$$

sq. units

Sol. 19. (a)

$$\text{Area} = \int_{\sqrt{3}}^2 \sqrt{4-x^2} dx = \left[\frac{x}{2} \sqrt{4-x^2} + \frac{4}{2} \sin^{-1} \frac{x}{2} \right]_{\sqrt{3}}^2$$

$$= \left[\frac{x}{2} \sqrt{4-x^2} + 2 \sin^{-1} \frac{x}{2} \right]_{\sqrt{3}}^2$$

$$\pi - \frac{\sqrt{3}}{2} - \frac{2\pi}{3} = \frac{\pi}{3} - \frac{\sqrt{3}}{2}$$

Sol. 20. (a)

$$\text{Area} = \int_0^{\sqrt{3}} \frac{x}{\sqrt{3}} dx + \int_{\sqrt{3}}^2 \sqrt{4-x^2} dx$$

$$= \frac{\sqrt{3}}{2} + \frac{\pi}{3} - \frac{\sqrt{3}}{2} = \frac{\pi}{3}$$

Sol. 21. (a)

$$\text{Area} = \int_0^{\pi/4} (\cos x - \sin x) dx = [\sin x + \cos x]_0^{\pi/4}$$

$$= \sqrt{2} - 1$$

Sol. 22. (a)

$$\text{Area} = \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx = [-\cos x - \sin x]_{\pi/4}^{\pi/2}$$

$$= \sqrt{2} - 1$$

Sol. 23. (d)

$$\sqrt{x} + \sqrt{y} = 1$$

$$\text{then } \sqrt{y} = 1 - \sqrt{x}$$

$$y = 1 + x - 2\sqrt{x}$$

$$\text{Area} = \int_0^1 (1 + x - 2\sqrt{x}) dx$$

$$\left[x + \frac{x^2}{2} - 2 \cdot \frac{x^{3/2}}{3/2} \right]_0^1 = 1 + \frac{1}{2} - \frac{4}{3} = \frac{1}{6}$$

Sol. 24. (c)

$$\text{if } x = 2$$

this line intersects with $y = |x-1|$ at (2,1)

$$\text{if } x = -2$$

this line intersects with $y = |x-1|$ at (-2,3)

Sol. 25. (c)

$$\text{Area} = \left[\int_{-2}^1 -(x-1) dx + \int_1^2 (x-1) dx \right]$$

$$= \left[-\frac{x^2}{2} + x \right]_{-2}^1 + \left[\frac{x^2}{2} - x \right]_1^2 = \frac{9}{2} + \frac{1}{2} = 5$$

Sol. 26. (c)

$$\text{for } x > 0$$

$$y = x^2 - 1 \text{ and } y = \frac{3}{2}x \text{ intersect at } (2,3)$$

$$\text{for } x < 0$$

$$y = -x^2 - 1 \text{ and } y = 2x \text{ intersect at } (-1, -2)$$

Sol. 27. (b)

$$\text{Area} = \left| \int_{-1}^0 (-x^2 - 1) dx \right| - \left| \int_{-1}^0 2x dx \right| + \left| \int_0^1 (x^2 - 1) dx \right|$$

$$+ \left| \int_0^2 \frac{3}{2}x dx \right| - \left| \int_1^2 (x^2 - 1) dx \right|$$

$$\left[-\frac{x^3}{3} - x \right]_{-1}^0 - \left[x^2 \right]_{-1}^0 + \left[\frac{x^3}{3} - x \right]_0^1 + \left[\frac{3x^2}{4} \right]_0^2 - \left[\frac{x^3}{3} - x \right]_1^2$$

$$= \frac{4}{3} - 1 + \frac{2}{3} + 3 - \frac{4}{3} = \frac{8}{3}$$

Sol. 28. (a)

$$\text{Area} = \int_{\frac{1}{2}}^1 f(x) dx = \int_{\frac{1}{2}}^1 (1 - x + x^2) dx$$

$$= \left[x - \frac{x^2}{2} + \frac{x^3}{3} \right]_{\frac{1}{2}}^1$$

$$= \left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{2} + \frac{1}{8} - \frac{1}{24} \right]$$

$$= \left[\frac{1}{3} + \frac{1}{8} - \frac{1}{24} \right]$$

$$= \frac{8+3-1}{24} = \frac{10}{24} = \frac{5}{12} \text{ sq. unit}$$

Sol. 29. (d)

$$\text{Area} = \int_1^{3/2} (x^2 + x - 1) dx$$

$$= \left[\frac{x^3}{3} + \frac{x^2}{2} - x \right]_1^{3/2}$$

$$= \left[\frac{27}{24} + \frac{9}{8} - \frac{3}{2} - \frac{1}{2} + 1 \right]$$

$$= \left[\frac{27+27-36-8-12+24}{24} \right]$$

$$= \left[\frac{78-56}{24} \right] = \left[\frac{22}{24} \right] = \frac{11}{12} \text{ sq. unit}$$

Sol. 30. (b)

$$|y| = 1 - x^2$$

if $y > 0$ then $y = 1 - x^2$ (downward parabola vertex at (0,1))

if $y < 0$ then $-y = 1 - x^2$ (upward parabola vertex at (0,-1))

$$\text{Area} = 2 \int_{-1}^1 (1 - x^2) dx$$

$$= 2 \left[x - \frac{x^3}{3} \right]_{-1}^1 = 2 \left[\left(1 - \frac{1}{3} \right) - \left(-1 - \frac{(-1)}{3} \right) \right] = \frac{8}{3}$$

Sol. 31. (c)

$$|x| + |y| = 1$$

this equation is the combination of four straight lines

$$x + y = 1$$

$$x - y = 1$$

$$-x + y = 1$$

$$-x - y = 1$$

and these equations make a square of side $\sqrt{2}$ so area of square = 2

Sol. 32. (c)

Both curves intersect at $x = 2$

$$\text{Area} = 2 \left[\int_0^2 \sqrt{3x} dx - \int_1^2 \sqrt{6(x-1)} dx \right]$$

$$= 2 \left[\left[\sqrt{3} \frac{x^{3/2}}{3/2} \right]_0^2 - \left[\sqrt{6} \frac{(x-1)^{3/2}}{3/2} \right]_1^2 \right]$$

$$= 2 \left[\left[\sqrt{3} \frac{2^{3/2}}{3/2} \right] - \left[\sqrt{6} \frac{(2-1)^{3/2}}{3/2} \right] \right]$$

$$= 2 \left[\frac{4\sqrt{6}}{3} - \frac{2\sqrt{6}}{3} \right] = \frac{4\sqrt{6}}{3}$$

Sol. 33. (b)

$$\text{Area} = \int_0^{\pi} c \sin x dx = c \int_0^{\pi} \sin x dx$$

$$= c [-\cos x]_0^{\pi} = c [1+1] = 2c$$

Sol. 34. (b)

It is a rectangle of dimension 10×8

$$\text{Area} = 10 \times 8 = 80 \text{ sq unit}$$

Sol. 35. (a)

Intersection point of both curve

$$y = x \text{ and } y^2 = 2x \text{ is } (0,0) \text{ and } (2,2)$$

$$\text{Area} = \int_0^2 (\sqrt{2x} - x) dx$$

$$= \left[\sqrt{2} \frac{x^{3/2}}{3/2} - \frac{x^2}{2} \right]_0^2 = \left[\sqrt{2} \frac{2^{3/2}}{3/2} - \frac{2^2}{2} \right]$$

$$= \left[\frac{8}{3} - 2 \right] = \frac{2}{3}$$

Sol. 36. (c)

Intersection point of both curve

$$y = x \text{ and } y^2 = 2x \text{ is } (0,0) \text{ and } (2,2)$$

$$\text{Area} = \int_0^2 (\sqrt{2x} - x) dx$$

$$= \left[\sqrt{2} \frac{x^{3/2}}{3/2} - \frac{x^2}{2} \right]_0^2 = \left[\sqrt{2} \frac{2^{3/2}}{3/2} - \frac{2^2}{2} \right]$$

$$= \left[\frac{8}{3} - 2 \right] = \frac{2}{3}$$

Sol. 37. (b)

$$\text{area of } x^2 + y^2 = 16 \text{ is } 16\pi$$

$$\text{then area of semicircle } y = \sqrt{16 - x^2} \text{ is } 8\pi$$

Sol. 38. (c)

$$\text{Between } x = -1.5 \text{ and } x = -1.8$$

$$[x] = -2$$

Area Bounded by x

$$\text{Axis} = \text{Area of Rectangle} = \text{Base} \times \text{Height}$$

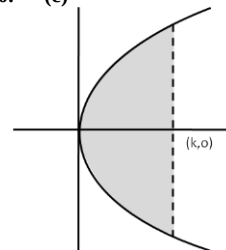
$$= 0.3 \times (2) = 0.6 \text{ square unit}$$

Sol. 39. (b)

$$\int_0^1 (x - x^3) dx$$

$$= \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

Sol. 40. (c)



$$2 \int_0^k \sqrt{4kx} dx = 24$$

$$\int_0^k \sqrt{4kx} dx = 24$$

$$\sqrt{k} \left[\frac{x^{3/2}}{3/2} \right]_0^k = 6$$

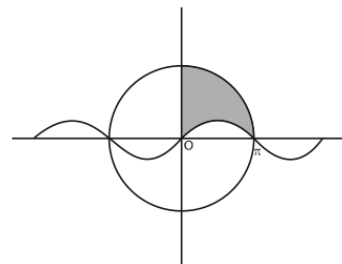
$$K^2 \times \frac{2}{3} = 6$$

$$K = 3.$$

Sol. 41. (b)

$$x^2 + y^2 = \pi^2$$

$$y = \sin x$$



$$\text{Area} = \int_0^{\pi} (\sqrt{\pi^2 - x^2} - \sin x) dx$$

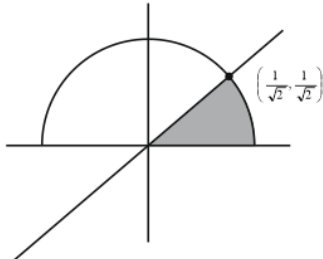
$$\Rightarrow \left[\frac{x}{2} \sqrt{\pi^2 - x^2} + \frac{\pi^2}{2} \sin^{-1} \frac{x}{\pi} + \cos x \right]_0^{\pi}$$

$$\Rightarrow \left(0 + \frac{\pi^2}{2} \sin^{-1}(1) - 1 \right) - (0 + 0 + 1)$$

$$\Rightarrow \frac{\pi^2}{4} - 2$$

Sol. 42. (c)

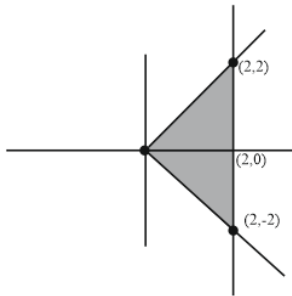
$$y = \sqrt{1 - x^2}$$



Area of complete circle is $\pi(1)^2 = \pi$
Area of $1/8^{\text{th}}$ portion of circle is $\pi/8$.

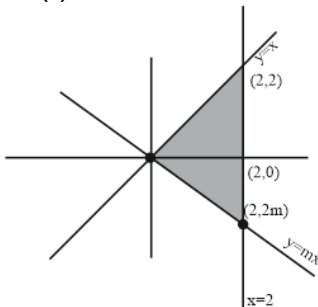
Sol. 43. (c)

$$|y| = x$$



Area given triangle is $1/2 \times 4 \times 2 = 4$ unit

Sol. 44. (a)



$$\text{Area} = \frac{1}{2} \times \text{Base} \times \text{Height} = 3$$

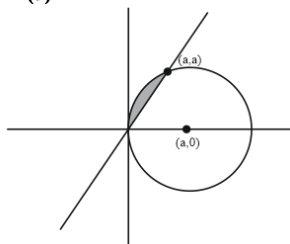
$$= \frac{1}{2} \times (2-2m) \times 2 = 3$$

We took here negative m, because m is given because m is given negative, but length is always positive

$$2-2m=3$$

$$m = -1/2$$

Sol. 45. (a)



Area of minor segment
= Area of quarter circle - Area of triangle
= $1/4 \pi a^2 - 1/2 \times a \times a$

$$= \left(\frac{\pi - 2}{4} \right) a^2$$

Sol. 46. (b)

In above graph area of major segment = Area of circle - area of minor segment

$$= \pi a^2 - \left(\frac{\pi - 2}{4} \right) a^2 = \left(\frac{3\pi + 2}{4} \right) a^2$$

Sol. 47. (a)

$$f(x) = x^2 \text{ for } x > 0$$

$$f(x) = -x^2 \text{ for } x > 0$$

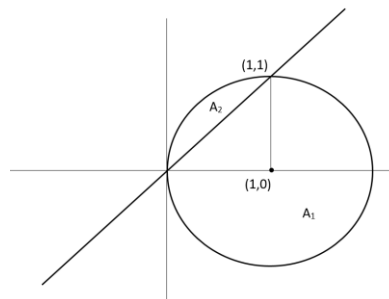
Area with x axis

$$= \int_{-1}^0 (-x^2) dx + \int_0^1 x^2 dx = 2 \left[\frac{x^3}{3} \right]_0^1 = \frac{2}{3}$$

Sol. 48. (d)

intersection points of line $y = x$ and circle $x^2 + y^2 - 2x = 0$ is (0,0) and (1,1)

Area of major segment



$A_1 = 3/4$ of area of circle + area of triangle

$$\frac{3}{4} \times \pi \times (1)^2 + \frac{1}{2} \times 1 \times 1 = \frac{(3\pi + 2)}{4}$$

Sol. 49. (a)

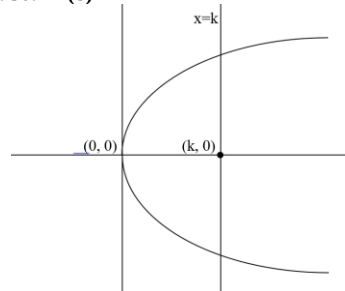
$$A_1 = \frac{(3\pi + 2)}{4}$$

$$A_2 = \text{area of circle} - A_1$$

$$A_2 = \pi(1)^2 - \frac{(3\pi + 2)}{4} = \frac{\pi - 2}{4}$$

$$\frac{2(A_1 + A_2)}{A_1 - 3A_2} = \frac{2\pi}{\left(\frac{3\pi + 2}{4} \right) - 3 \left(\frac{\pi - 2}{4} \right)} = \pi$$

Sol. 50. (b)



Area of shaded portion =

$$2 \int_0^k \sqrt{kx} dx = \frac{4}{3}$$

$$2\sqrt{k} \left[\frac{x^{3/2}}{3/2} \right]_0^k = \frac{4}{3}$$

$$\sqrt{k} \left(\frac{k^{3/2}}{3/2} \right) = \frac{2}{3}$$

$$\left(\frac{2k^2}{3} \right) = \frac{2}{3}$$

$$k^2 = 1$$

$$k = 1$$

Sol. 51. (a)

by above solution $k = 1$

so equation of parabola $y^2 = x$

focus of this parabola is at $\left(\frac{1}{4}, 0 \right)$

$$\text{area with latus rectum} = 2 \int_0^{\frac{1}{4}} \sqrt{x} dx$$

$$2 \left[\frac{x\sqrt{x}}{3/2} \right]_0^{\frac{1}{4}} = \frac{1}{6}$$

Sol. 52. (c)

By the concept of differential equation

$$\text{slope} = \frac{dy}{dx} = 4$$

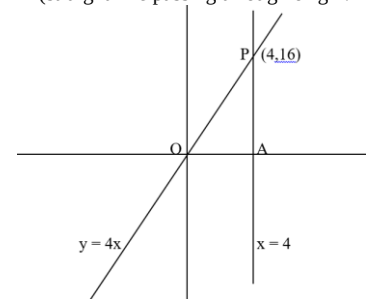
$$dy = 4dx$$

$$\int dy = \int 4dx$$

$$y = 4x + c$$

passing through origin so $c = 0$

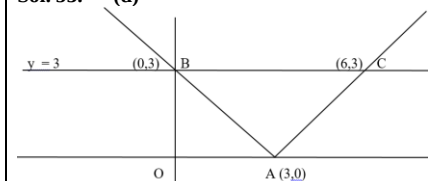
$y = 4x$ (straight line passing through origin.)



Area of triangle OPA

$$= \frac{1}{2} \times 4 \times 16 = 32 \text{ sq. unit}$$

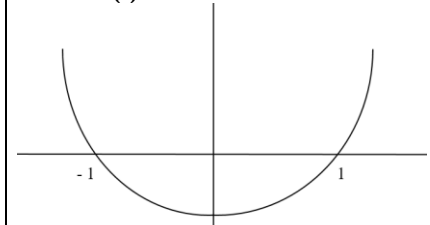
Sol. 53. (d)



Area of triangle ABC

$$= \frac{1}{2} \times 6 \times 3 = 9 \text{ sq. unit}$$

Sol. 54. (c)



Area bounded by parabola with x axis

$$= \int_{-1}^1 (x^2 - 1) dx = \left[\frac{x^3}{3} - x \right]_{-1}^1 = \frac{4}{3}$$

Sol. 55. (d)

$$f(x) = x^2 \text{ for } x > 0$$

$$f(x) = -x^2 \text{ for } x > 0$$

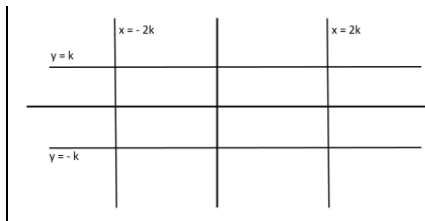
Area with x axis

$$= \left| \int_{-2}^0 (-x^2) dx \right| + \int_0^1 x^2 dx$$

$$= \left[-\frac{x^3}{3} \right]_{-2}^0 + \left[\frac{x^3}{3} \right]_0^1 = \frac{8}{3} + \frac{1}{3} = 3$$

Sol. 56. (d)

$-2k \leq x \leq 2k$ and $-k \leq y \leq k$,



Area of rectangle = length $\times$ breadth
 $= 4k \times 2k$